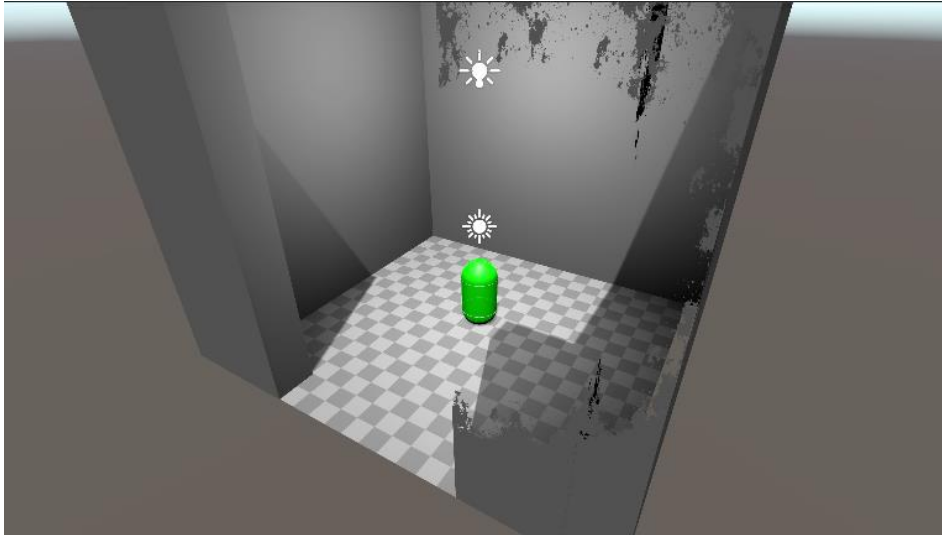


PixelPulse – Documentation



Dynamic Occlusion System

Script

This script is used to **calculate** and **update** the Dynamic Occlusion System (DOS) **Materials**.

It has the following parameters:

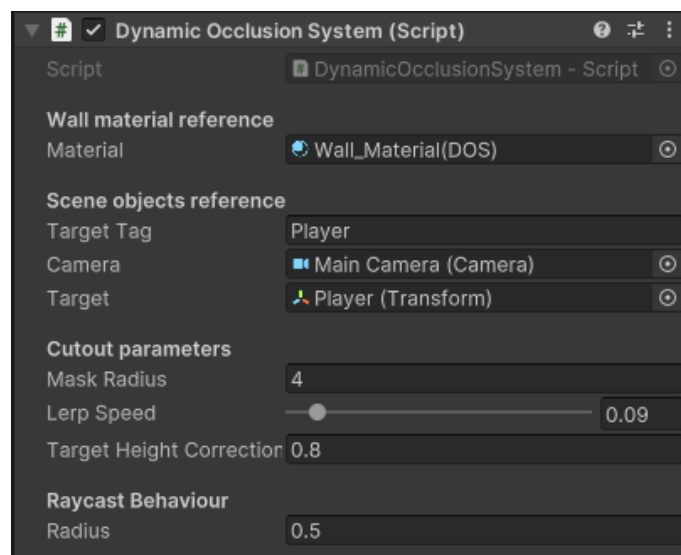


Fig. 1 – DOS script in Unity's Inspector view

- **Material** is the reference of the DOS material to update.
- **Target Tag** is the Tag the script will look for in `Awake()`, leave it empty if the target is manually assigned.
- **Camera** is the Camera on which the occlusion will be active.
- **Target** is the target behind the objects that will be cutout.

- **Mask Radius** is the radius of the cutout mask.

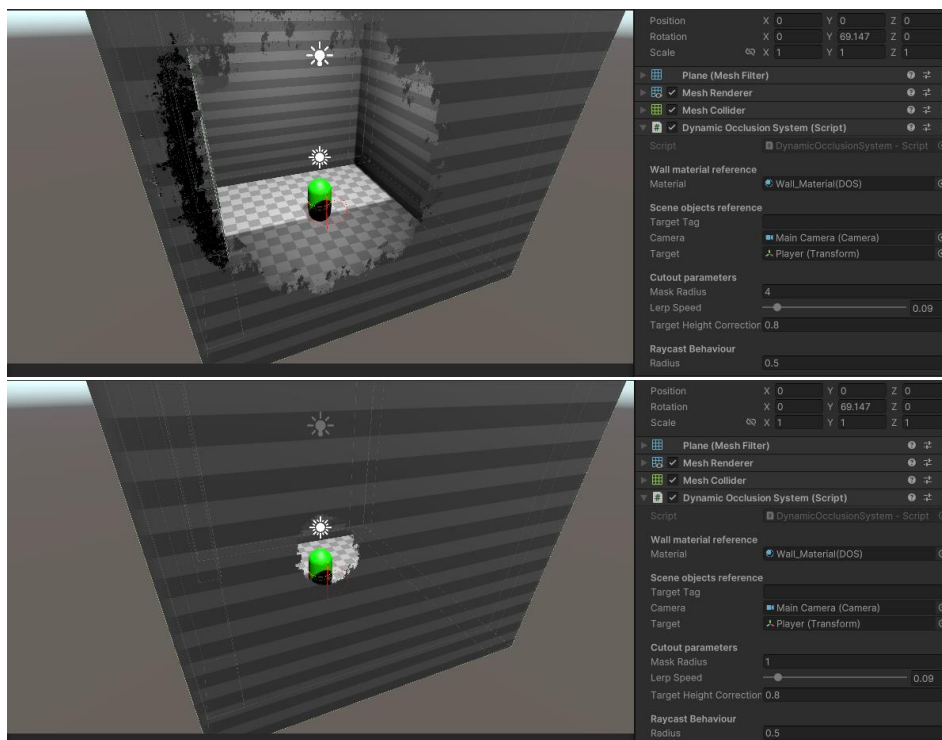


Fig. 2 - Mask Radius at 4.0f and 1.0f

- **Lerp Speed** will define the speed at which the cutout will update.
- **Target Height Correction** gives the possibility to change the position of the cutout on the Y Axis.

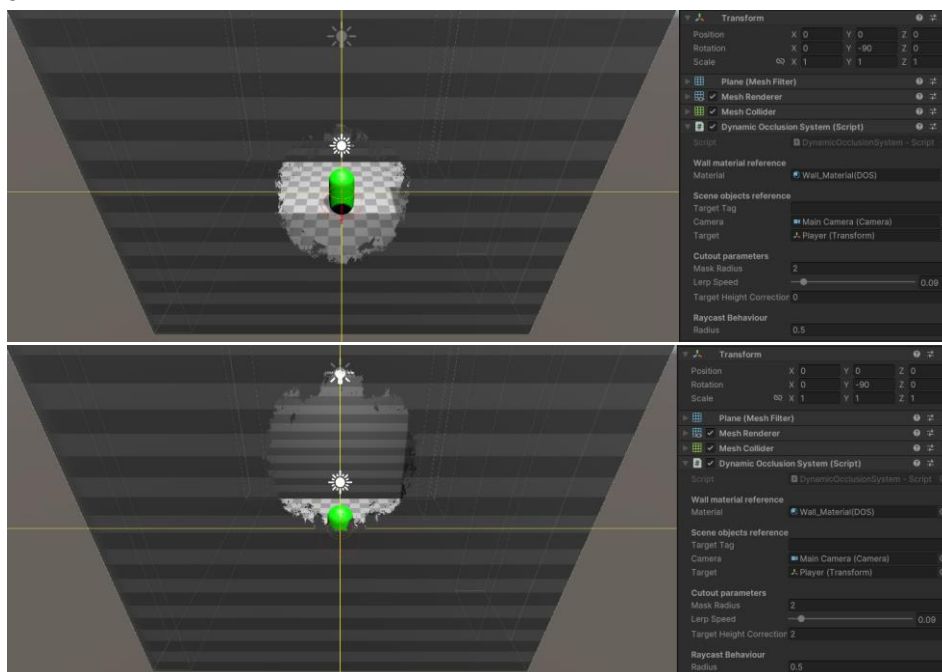


Fig. 3 - Target Height Correction set at 0.0f and 2.0f (exaggerated)

- **Radius** will change the size of the SphereCast, changing radius of wall detection.

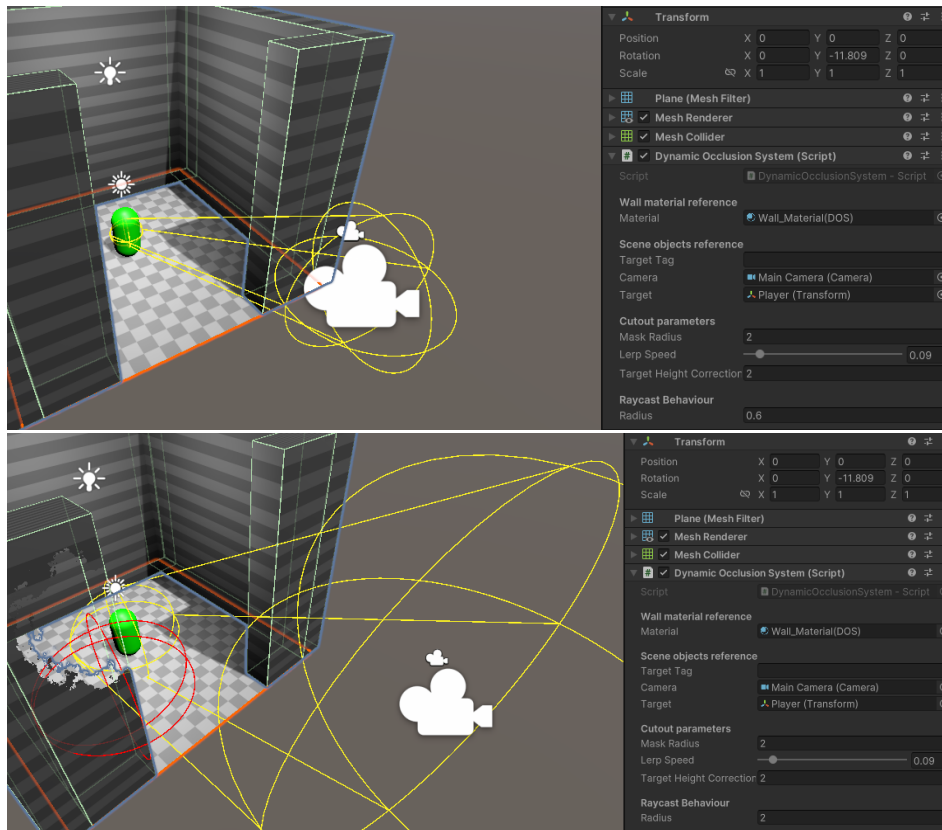
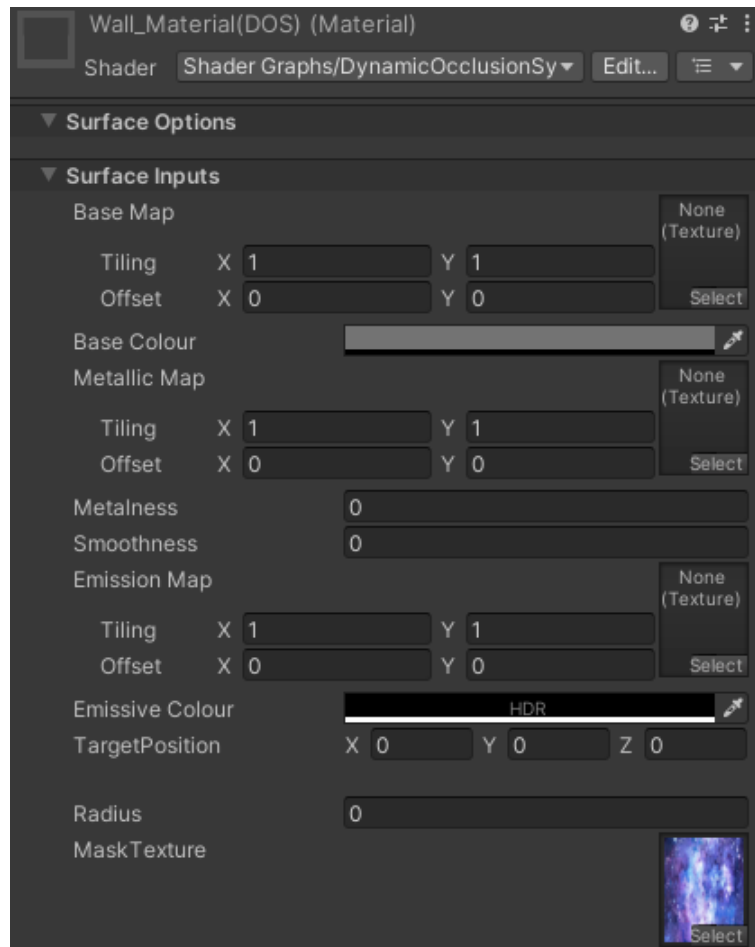


Fig. 4 – Radius of SphereCast set to 0.6f and 2.0f (exaggerated)

Shader

The shader enables the creation of DOS Materials.



This material has the original surface shader canals:

- **Base Map & Base Colour**
- **Metallic Map & Metalness**
- **Smoothness**
- **Emission Map & Emissive Colour**

And the DOS parameters:

- **Target Position** is updated by the script (can be changed manually for debugging in editor)
- **Radius** is updated by the script (can be changed manually for debugging in editor)
- **Mask Texture** is the texture used for the occlusion cutout.



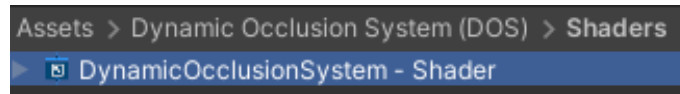
**MaskTexture can be changed in the Shader by a noise but
It originally uses TRIPLANAR UVs so be sure
YOU KNOW WHAT YOU ARE DOING BEFORE YOU CHANGE IT**



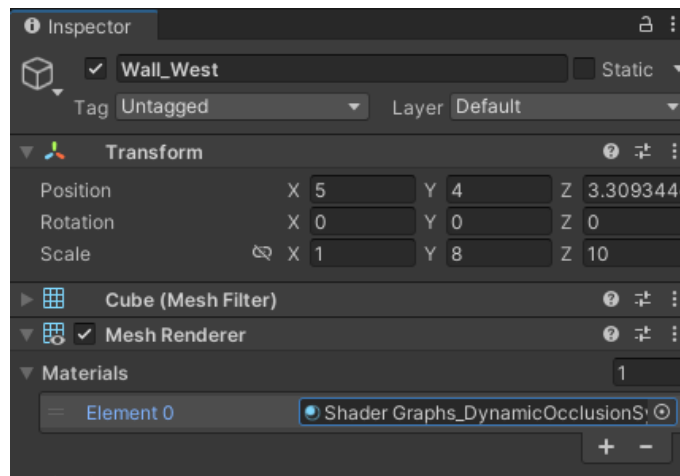
How to use it?

To use this system:

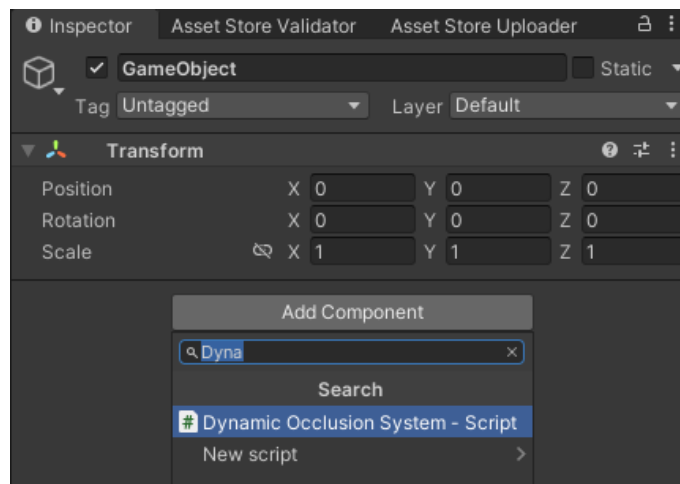
- **Create a material** derived from the **Dynamic Occlusion System - Shader**



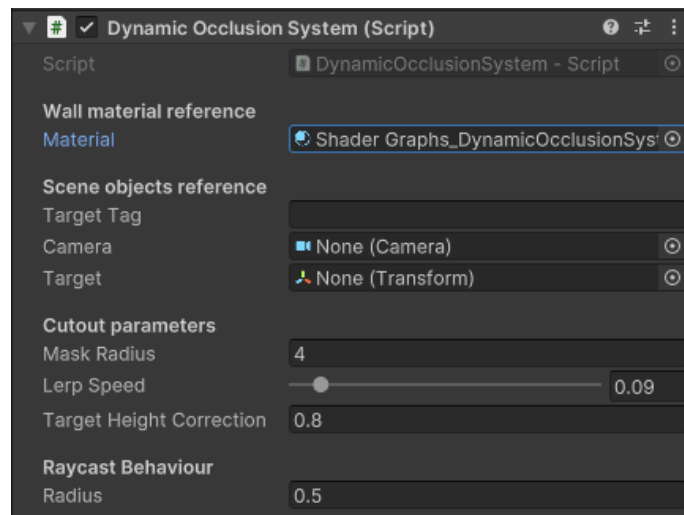
- **Right click** on the shader > **Create > Material**
In *Assets > Dynamic Occlusion System (DOS) > Shader*
- **Place the Material** on the GameObjects (ex. Walls) you want to see through
Inspector > Mesh Renderer > Materials > Element (n)



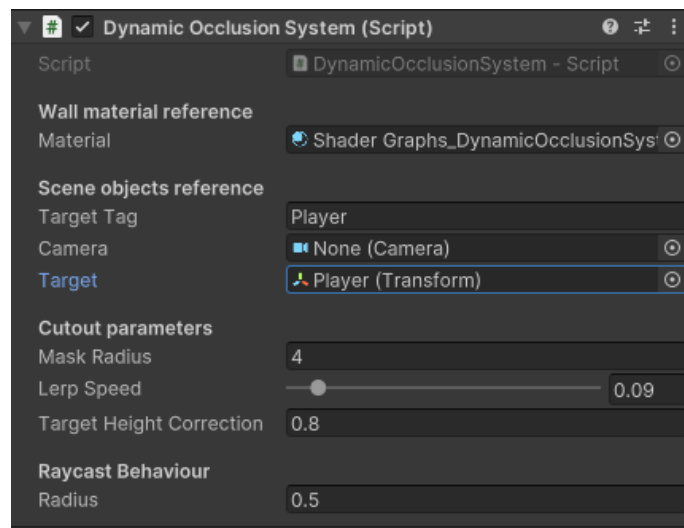
- **Assign the Dynamic Occlusion System - Script** on any object in your scene



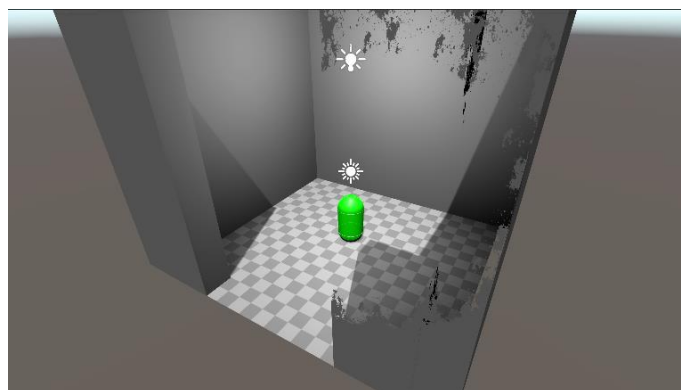
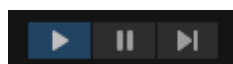
- **Assign** the newly created **material**



- Enter the **Target Tag** or Assign the **Target**



- Finally click **Play** and Enjoy 😊



The PixelPulse team thanks you for purchasing
Dynamic Occlusion System - DOS
and wishes you all the best for your project.
